Multivariate Data Analysis - MESIO UPC

Biplots

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Biplots

- Graphical representation technique of a matrix of dimensions $n \times p$.
- It is called **BI**plot because the rows and the columns are represented simultaneously, providing visual information about their relationships.
- It is a generalization of the scatterplot.

Biplot vs Scatterplot

- A **scatterplot** focuses on visualizing the relationship between two variables, whereas a **biplot** displays the relationships between both variables and observations simultaneously.
- Scatterplots are inherently two-dimensional. Biplots can represent relationships in higher dimensions on a two-dimensional plane through a **low-rank projection**.
- **Biplot axes** (representing variables) are often **oblique**, as their angles reflect the correlations between variables more accurately than fixed orthogonal axes.
- The graphical representation in a biplot is a **rank-k approximation** of the data, typically optimized using a least-squares criterion (SVD).
- Biplots facilitate the interpretation of complex patterns across multiple variables and observations at once, whereas scatterplots are generally limited to a simple x-y comparison.

Goal

The objective is to visualize and elucidate the relationships among variables as reflected in the sample data.

Building a Biplot

We need to factorize the data matrix into two different matrices.

$$\mathbf{X}_{n\times p} = \mathbf{F}_{(n\times k)}\mathbf{G}_{(k\times p)}$$

- **X** is called the target matrix.
- **F** Left matrix or **Row Markers** (in some literature are called "scores"). This matrix belongs to the vector space of the observations.
- **G** Right matrix or **Column Markers** (in some literature are called "loadings"). This matrix belongs to the vector space of the variables.
- Often row markers are represented as points and column markers as arrows (But is not mandatory to do so).

When $k = 2$ we can plot both matrices at once!

Dot product (again): Interpretation and geometry

- We will think of the rows of **F** as **points** and the columns of **G** as **vectors** (axes of the Biplot).
- We know that $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| cos(\mathbf{u}, \mathbf{v})$ so $\|\mathbf{p}\| = \|\mathbf{v}\| cos(\phi)$
- This enables the projection of the rows of **F** onto the columns of **G**,facilitating the interpretation of both individuals and variables.

Example

$$\begin{bmatrix} 8 & 2 & 2 & -6 \\ 5 & 0 & 3 & -4 \\ -2 & -3 & 3 & 1 \\ 2 & 3 & -3 & -1 \\ 4 & 6 & -6 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 2 \\ -1 & 1 \\ 1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & -1 & -2 \\ 1 & -1 & 2 & -1 \end{bmatrix}$$

Computing $\mathbf{F}$ and $\mathbf{G}$

- There exist infinitely many solutions for determining the involved matrices due to $\mathbf{X} = \mathbf{F}\mathbf{T}\mathbf{T}^{-1}\mathbf{G}$.
- Using optimization and the least squares criterion we can obtain an approximation in order to obtain rank 2 matrices.
- The *Eckart-Young Theorem* (SVD) gives us the answer.

Interpretation I

- It is important to control the **aspect ratio** of the biplot.
- The precise interpretation depends on the matrix that is represented and on the chosen scaling.
- In general, the **origin** represents the **mean vector**.
- **Orthogonal projections** of points (or vectors) onto vectors approximate the entries of the matrix of interest.
- **Distances** between points reflect some chosen measure of similarity of the corresponding observations.

Interpretation II

- **Vector Length**: A **longer axis** (vector) generally indicates a **larger variance** or a higher quality of representation for that variable within the displayed dimensions. **Standardization** ensures that all axes have equal variance and then **Vector Length** is only related to **quality of representation**.
- **Proximity and Correlation**: The **proximity of axes** (a small angle between them) signifies similar projection values for the observations, which implies a **strong positive correlation** between the associated variables. This proximity implies that observations will have similar projection values on both axes, facilitating a joint interpretation of individuals and variables.
- **Opposed Axes**: Axes pointing in **opposite directions** signify **strong negative correlations**. In this case, observations projecte onto these axes will reflect contrasting values for those variables.

Biplots and SVD

The singular value decomposition (SVD) yields multiple valid factorizations or ways to combine $\mathbf{F}$ and $\mathbf{G}$:

1. $\mathbf{F} = \mathbf{U}\boldsymbol{\Sigma}$ and $\mathbf{G} = \mathbf{V}$
2. $\mathbf{F} = \mathbf{U}$ and $\mathbf{G} = \mathbf{V}\boldsymbol{\Sigma}$
3. $\mathbf{F} = \mathbf{U}\boldsymbol{\Sigma}^{\frac{1}{2}}$ and $\mathbf{G} = \mathbf{V}\boldsymbol{\Sigma}^{\frac{1}{2}}$

In general, $\mathbf{F} = \mathbf{U}\boldsymbol{\Sigma}^{\alpha}$ and $\mathbf{G} = \mathbf{V}\boldsymbol{\Sigma}^{(1-\alpha)}$ with $\alpha \in [0, 1]$. Note that different scalings have **different geometrical properties**.

Biplots calibration

- **Calibration** refers to **drawing scales with tickmarks and labels** along the biplot vectors.
- It facilitates biplot interpretation and **data recovery**, enabling the user to estimate the original values of observations by projecting them onto the calibrated axes.
- While calibrated scales are **usually omitted**, they are included when a specific variable is of particular interest or when the biplot is used as a predictive tool.

Supplementary information

- Once a biplot has been made, additional observations or variables can be added by projection.
- Those observations are called **supplementary points or variables**.
- Coordinates for supplementary points/variables can be obtained by regression onto the biplot axes.

Supplementary information and regression

From

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \mathbf{F}\mathbf{G}^\top$$

it follows that

$$\mathbf{G} = \mathbf{X}^\top \mathbf{F}(\mathbf{F}^\top \mathbf{F})^{-1}$$

and

$$\mathbf{F} = \mathbf{X}\mathbf{G}(\mathbf{G}^\top \mathbf{G})^{-1}$$

- **Golden rule**: Row and column markers are **regression coefficients**.
- Substitute supplementary data for $\mathbf{X}$ to compute the markers.

Biplots and MVA methods (1)

- We can interpret SVD and biplots as performing multiple linear regression taking one of the vector variables as the target variable. Nearly all MVA techniques are based on SVD. This **automatically allows us** to use Biplots.

- MVA methods are defined by the matrix being approximated:
  - PCA: $\mathbf{X}$, $\mathbf{S}$ or $\mathbf{R}$.
  - CA: Contingency matrix or Burt table.
  - CCA: $\mathbf{R}_{xy}$.
  - MANOVA: Matrix of means.
  - ...

Biplots and MVA methods (2)

Matrix	Data	Method
$\mathbf{X}$	[raw data]	Principal component analysis (PCA)
		Scatter plot + regression
$\mathbf{X}_c$	[centered]	Principal component analysis (PCA)
		Scatter plot + regression
$\mathbf{X}_s$	[standardized]	Principal component analysis (PCA)
		Scatter plot + regression
$\mathbf{X}_{co}$	[compositional]	Log-ratio principal component analysis (LR-PCA)
$\mathbf{S}$	[covariances]	Principal component analysis (PCA)
$\mathbf{R}$	[correlations]	Principal component analysis (PCA)
		Factor analysis (FA)
$\mathbf{N}$	[cross table; profiles]	Correspondence analysis (CA)
$\mathbf{M}$	[table of group means]	Linear discriminant analysis (LDA)
$\mathbf{R}_{xy}$	[correlations between sets]	Canonical correlation analysis (CCA)
$\mathbf{B}_{xy}$	[regression coefficients]	Redundancy analysis (RDA)

References

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